

TABLE 1
SRAM Vs DRAM

Parameter	SRAM	DRAM
Read/write speed	Faster	Slower than SRAM
Power consumption	More	Less
Internal structure	Complex	Simpler than SRAM
Density	Less dense	Highly dense
Used in	Cache memory	Main memory
Storage per bit	Can store more bits per chip	Cannot store many bits per chip
Storage component	Transistor	Capacitor
Price	Expensive	Economical than SRAM

(Credit: networkinterview.com)

TABLE 2
SPECIFICATION OF DDR5 AND Lpddr5

Factor	DDR5	Lpddr5
Maximum die density	64Gbit	32Gbit
Max. data rate	6.4Gbps	6.4Gbps
Max. UDIMM size (DSDR)	128GB	N/A
Channels	2	1
Channel width	64 bits (2×32)	16 bits
Memory banks	4	16
Bank groups	8/4	4
Burst length	BL 16×2	BL 16
Prefetch	16n	16n
Voltage(Vdd)	1.1V	1.05V
Vddq	1V	0.5V

(Credit: hardwaretimes.com)

on the temperature range where DRAM should operate. Data integrity is lost at a level greater than a certain temperature.

Loss of power. DRAM may lose its contents due to a loss of power. So, a backup non-volatile memory is required where DRAM contents need to be copied before the power is lost. When power is regained, stored contents in non-volatile memory are written back to DRAM.

Specification of DDR5 and Lpddr5 as set by Joint Electron Device Engineering Council (JEDEC), which is an industry standards body and governs specifications of DRAMs, are given in Table 2.

DDR5 MC and PHY

DDR5 is described as the fifth-generation Double Data Rate Synchronous DRAM designed for next-generation computing. DDR5 Memory Controller (MC) and DDR PHY are two major components for efficient DRAM performance. DDR5 MC needs digital design experts while DDR5 PHY needs both digital and analogue expertise. For each application, DDR5 has a large number of timing parameters and configuration registers. Each vendor has to own its own unique timing parameter and configuration register values. DDR5 PHY Interface (DFI) is an interface protocol that defines timing, signals, and programmable parameters required to transfer data to and from DRAM to MC. In many situations, MC and PHY are designed separately and often by different companies.

Next generation DRAMs

Current DRAMs increase latency and power consumption, known as the 'memory wall.' Next-generation DRAMs have come up with new features to overcome these challenges. Introduced by Intel in 2015, 3D XPoint is described as the next generation DRAM. 3D XPoint is based on a technology called Phase-Change Memory (PCM) and is used in SSDs and DIMMs. PCM

stores information in the amorphous and crystalline phases.

Meanwhile, the industry is also developing other new memory types, such as Magnetoresistive RAM (MRAM) and Resistive RAM (ReRAM). Like 3D XPoint, MRAM and ReRAM can be made and sold as standalone devices.

3D XPoint is not sold as an embedded memory. In contrast, MRAM and ReRAM can also serve in the embedded memory markets. For MRAM, the industry is developing a next-generation technology called Spin-Transfer Torque MRAM (STT-MRAM). STT-MRAM uses the magnetism of electron spin to provide non-volatile properties in chips, combining the speed of SRAM and the non-volatility of flash with unlimited endurance.

In traditional memory, data is stored as an electric charge. In contrast, MRAM uses a magnetic tunnel junction (MTJ) memory cell for the storage element. Ferroelectric RAM (FRAM) is another technology to watch. Using a ferroelectric capacitor to store data, FRAM is a non-volatile memory with large storage capacity. Nantero, a private company, has more than 170 nanotube patents to replace DRAMs. It is possible to have hundreds and thousands of such nanotubes in every memory cell to form a network of resistive elements to form non-volatile random access memory (NVRAM). Undoubtedly, next-generation DRAMs are providing plenty of opportunities for OEMs.

Our past, present, and future memory requirements in the e-world are much more than we expect. In the web world, information can be kept alive by memory. DRAM gives another view of e-memory, providing another understanding of the 'data storage mechanism.' **EFY**